



Flip Flop

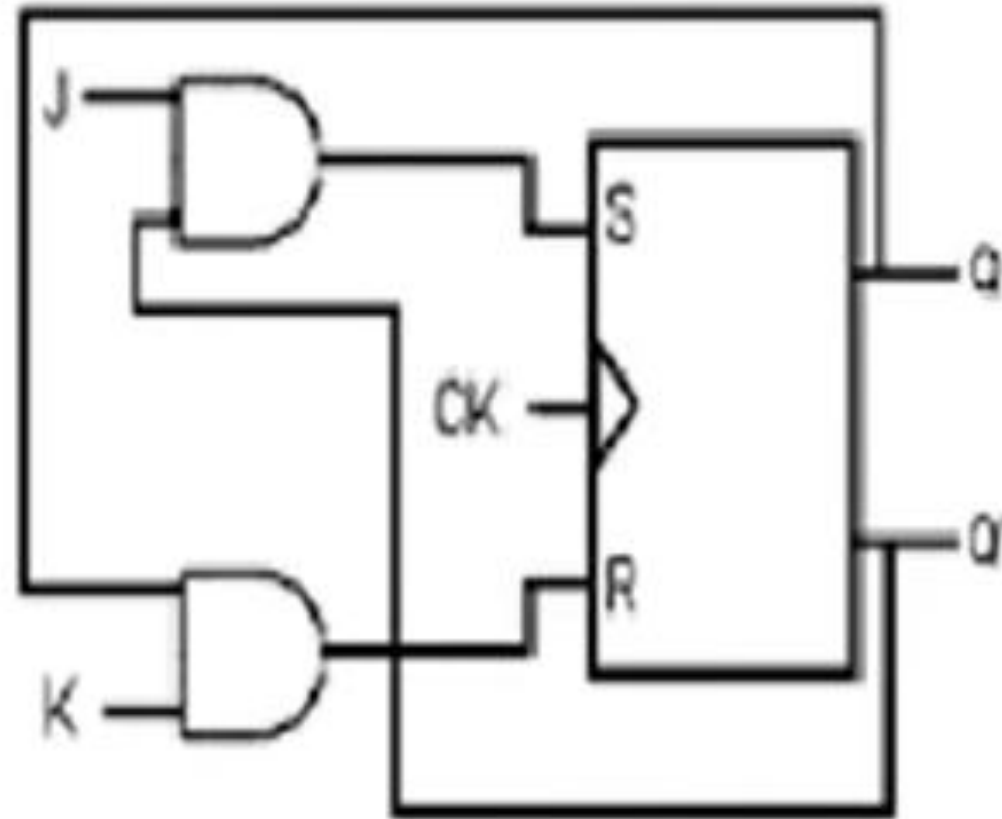
Agenda

- Código em sistemas digitais
- Codificadores
- Decodificadores
- Latch
- Flip Flop RS

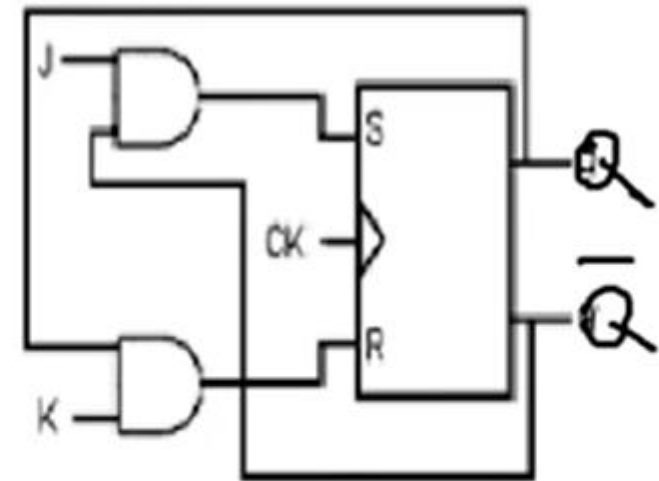
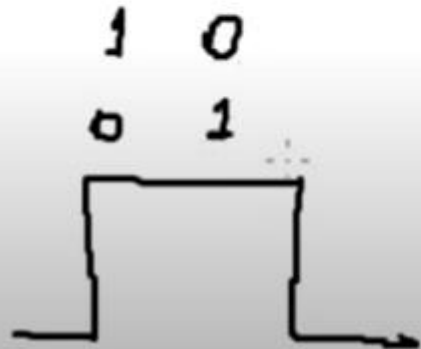
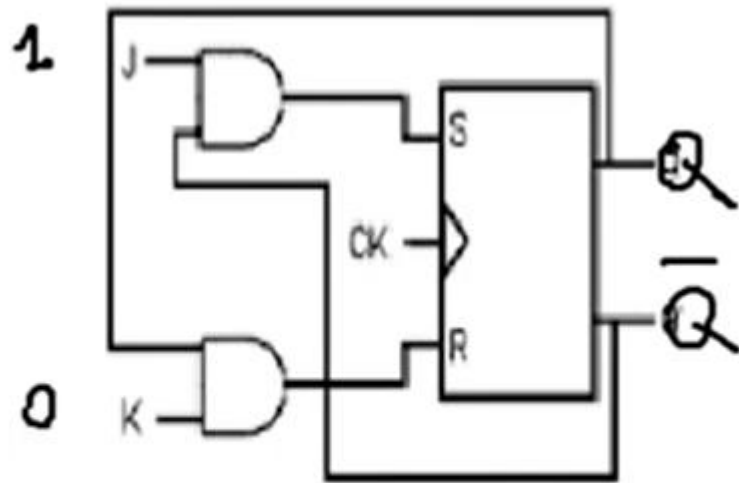


Flip Flop JK

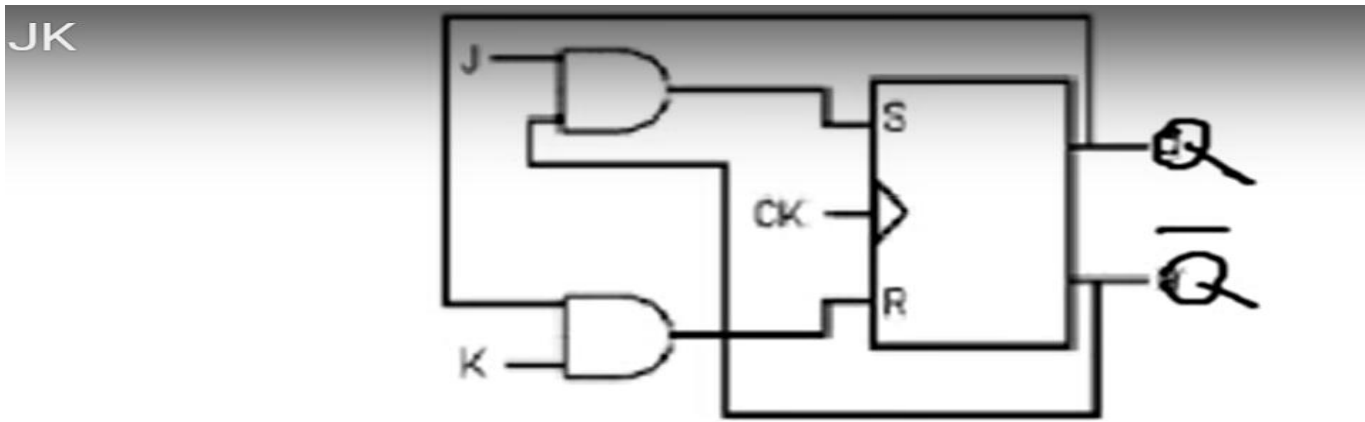
- Anteriormente
 - Mudança assíncrona
 - Latch
 - Mudança síncrona
 - Flip Flop RS



Problema do clock



Estado Proibido



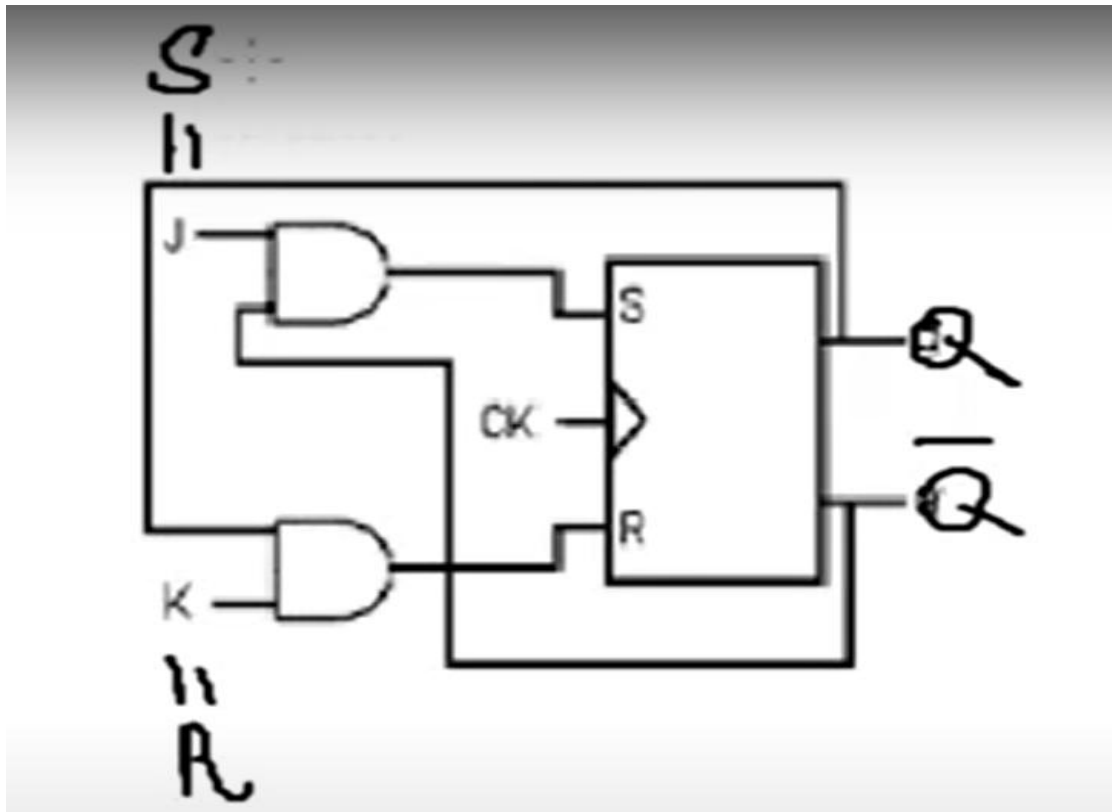
FLIP FLOP JK – Resolve o problema do Estado proibido

$$Q = \bar{Q} = 0$$

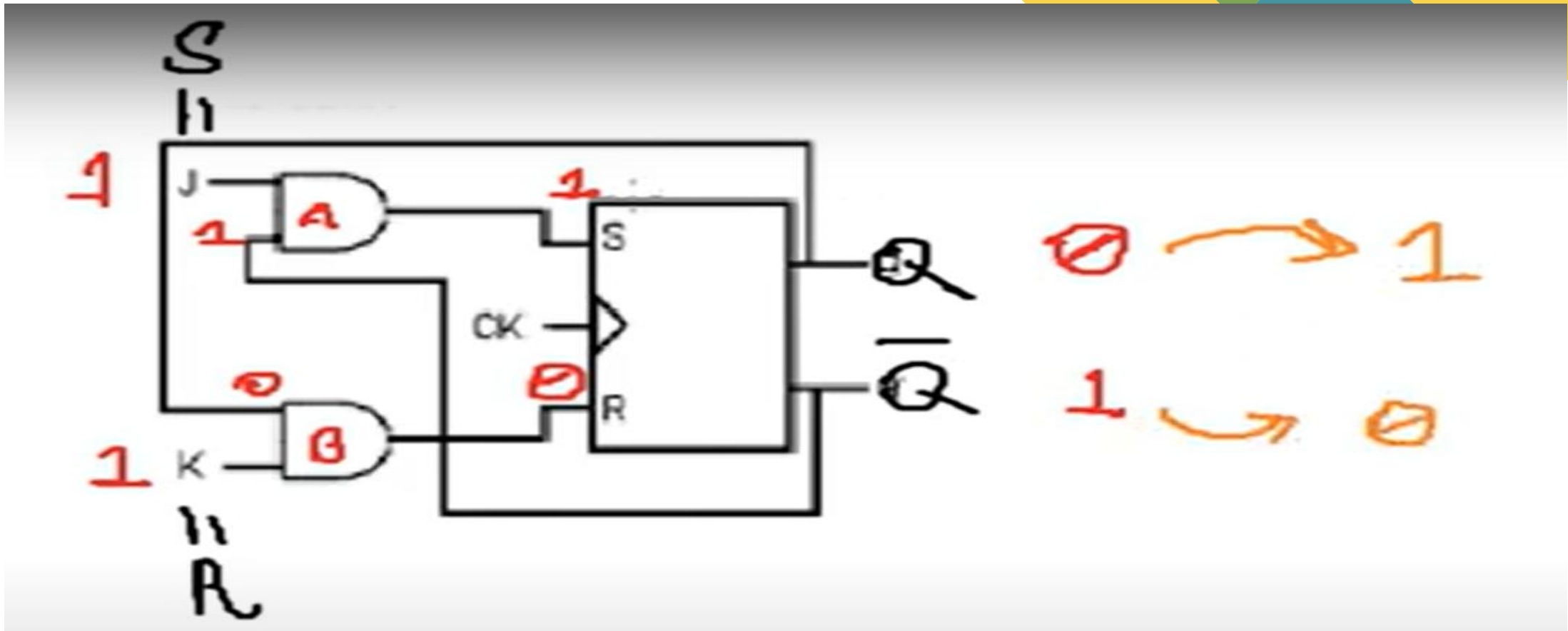
Ou

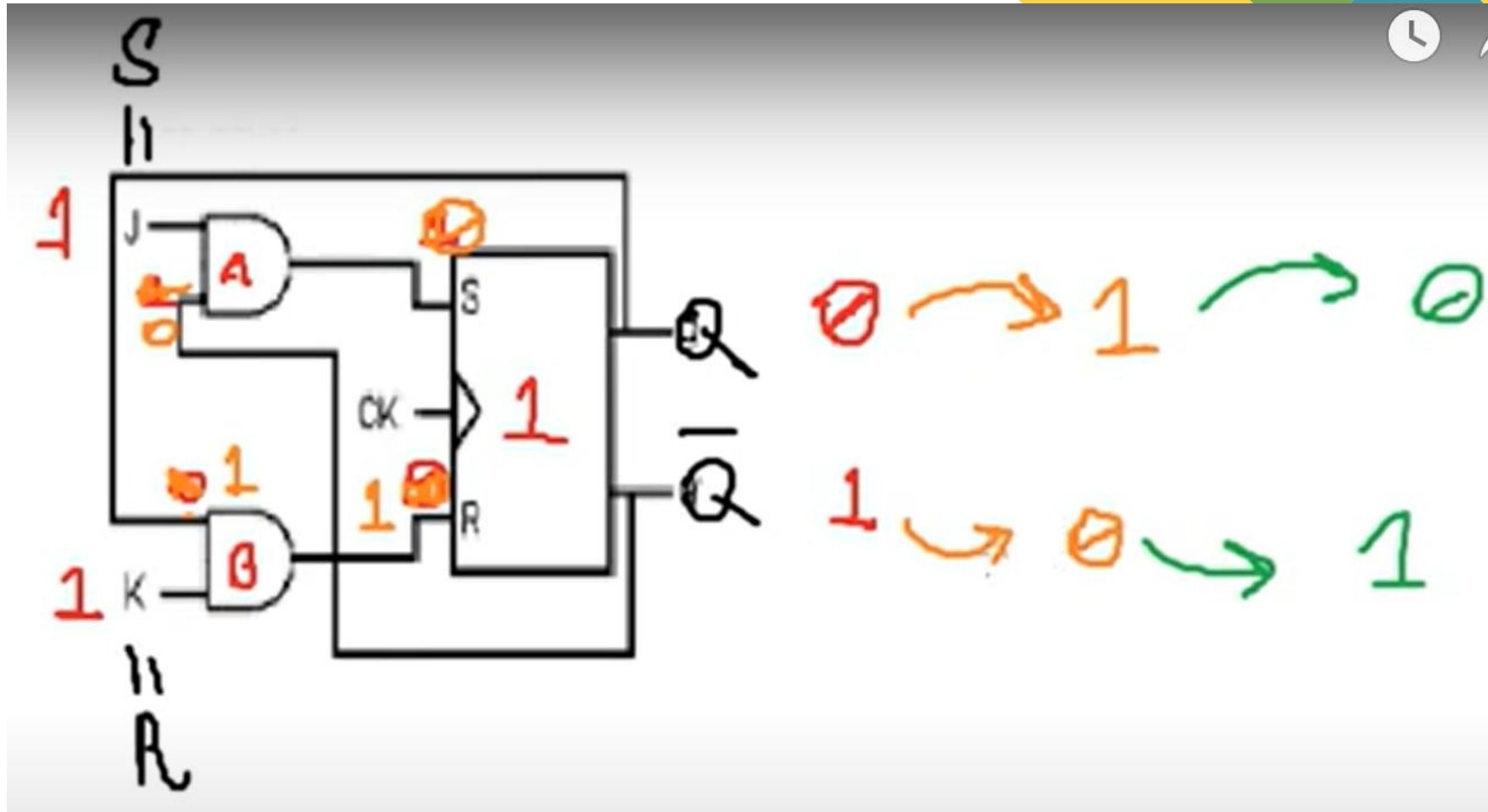
$$Q = \bar{Q} = 1$$

Analogo Flip Flop RS

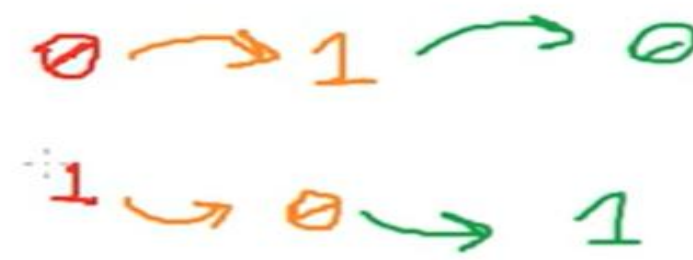
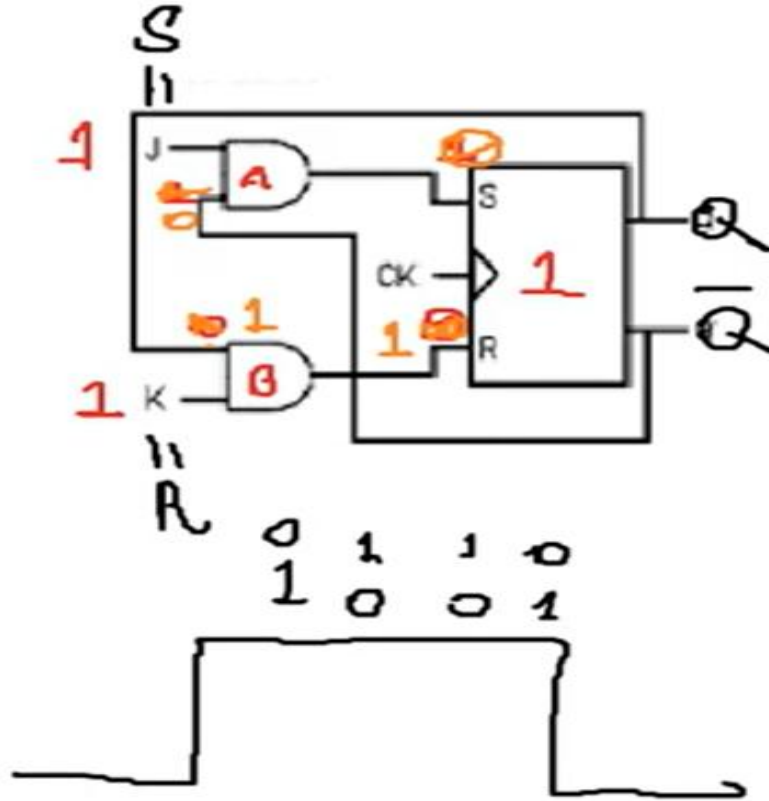


$J=1, K=1 \Rightarrow R=S=1$





Estado Proíbido



Detector de borda

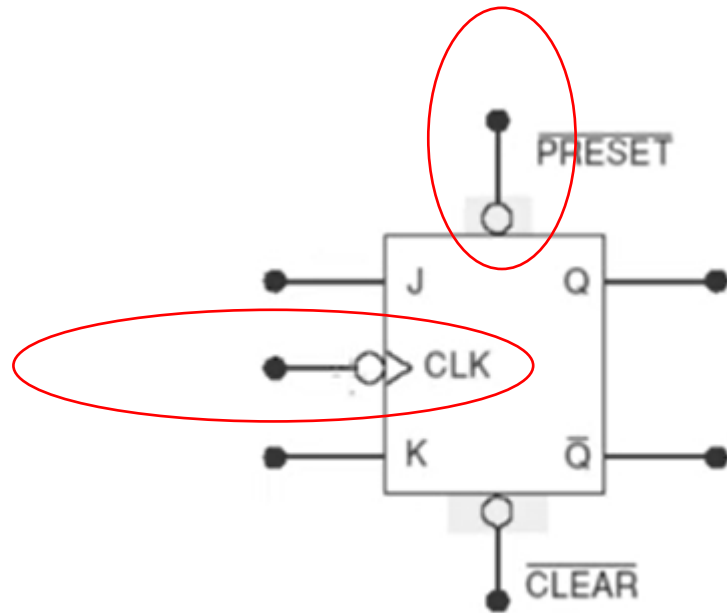
Flip Flop JK



J	K	Q
0	0	Q
1	0	1
0	1	0
1	1	Q-bar

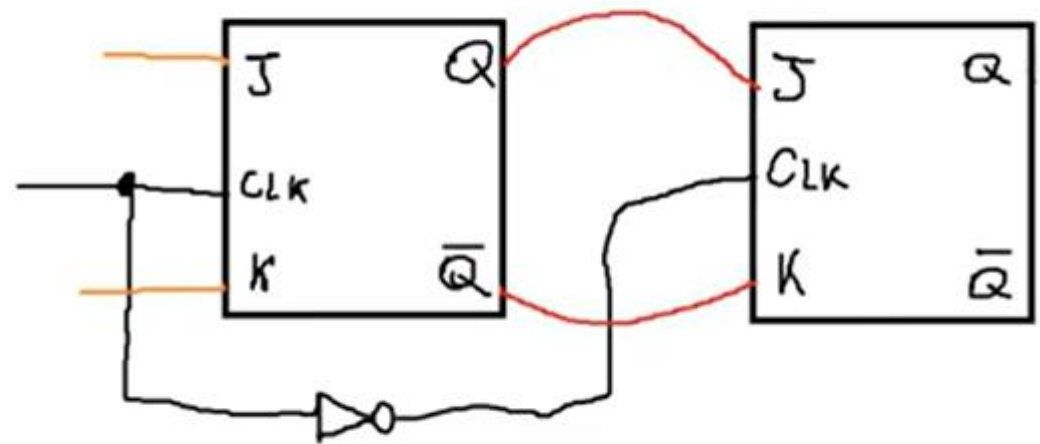


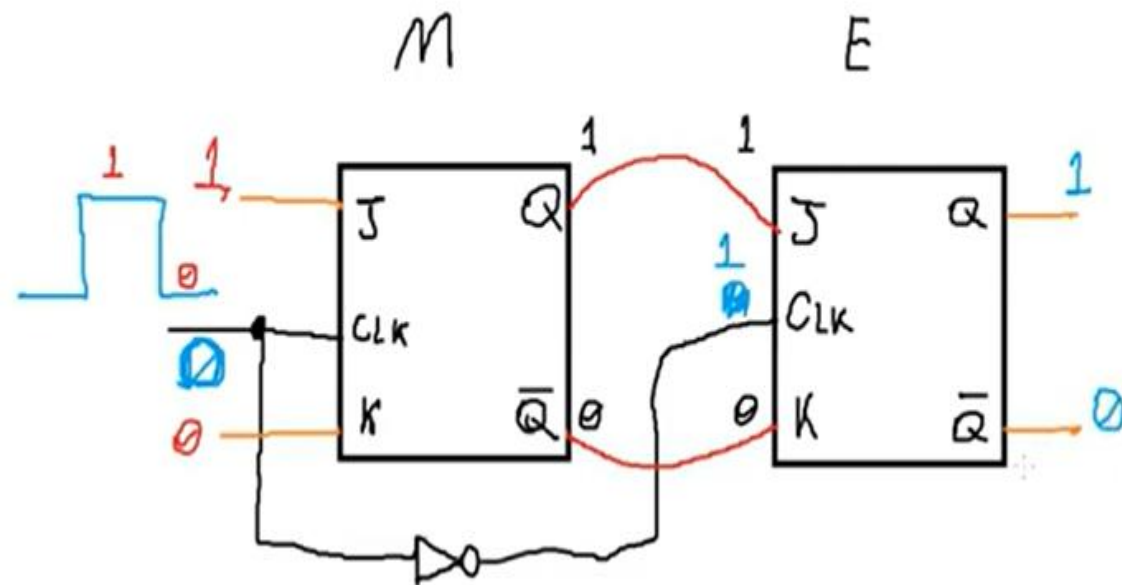
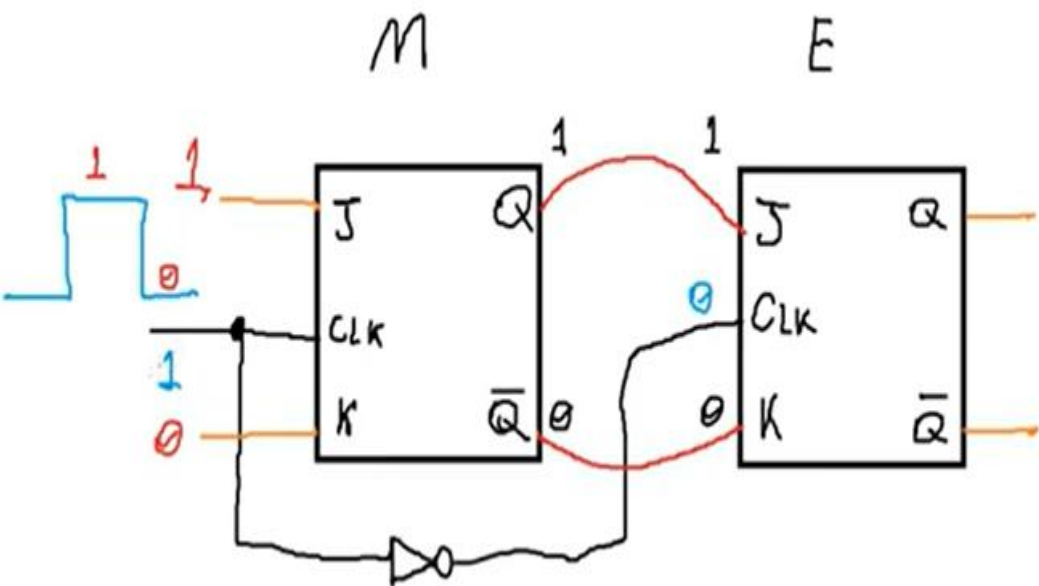
Datasheet



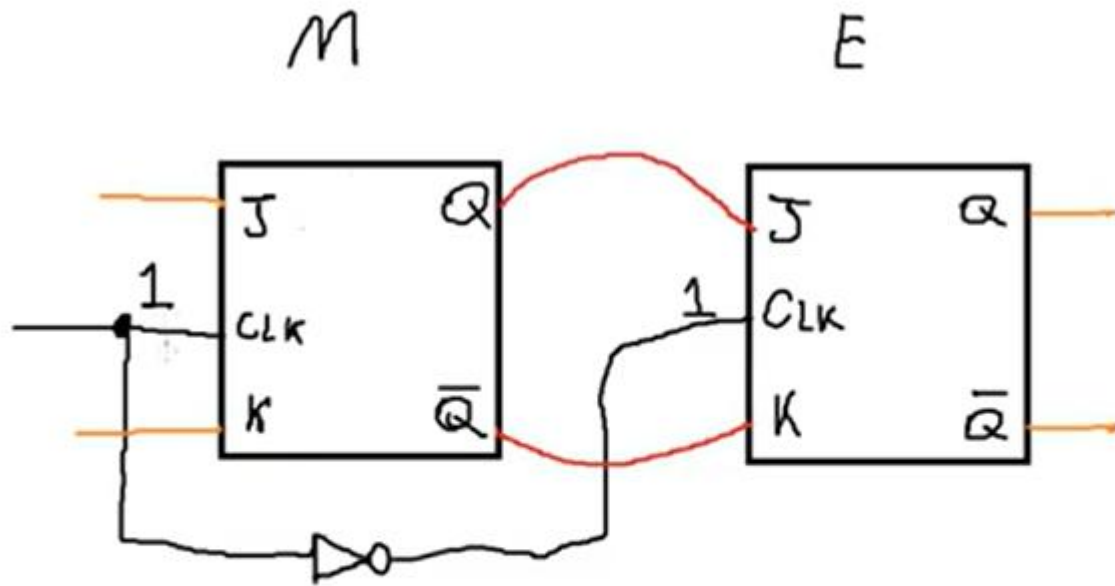
Flip Flop Mestre Escravo

- Não havia tecnologia para fazer chaveamento por borda
- Pode ser flip flop RS





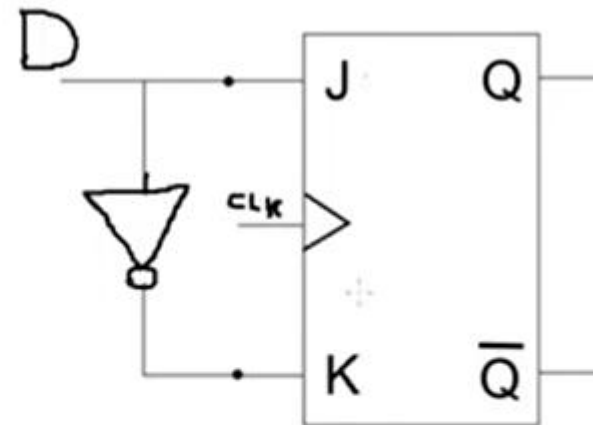
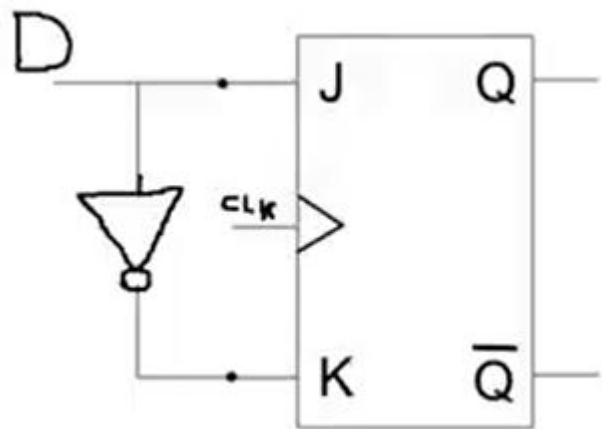
Funcionamento simultâneo



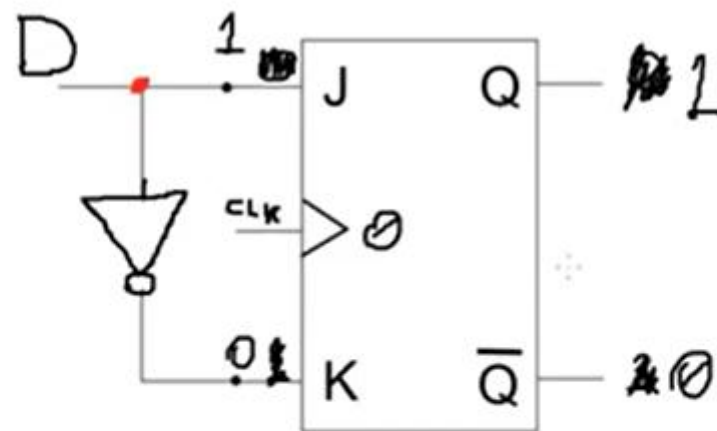
Aumenta Frequência

Sensível a entrada

Flip Flop tipo D

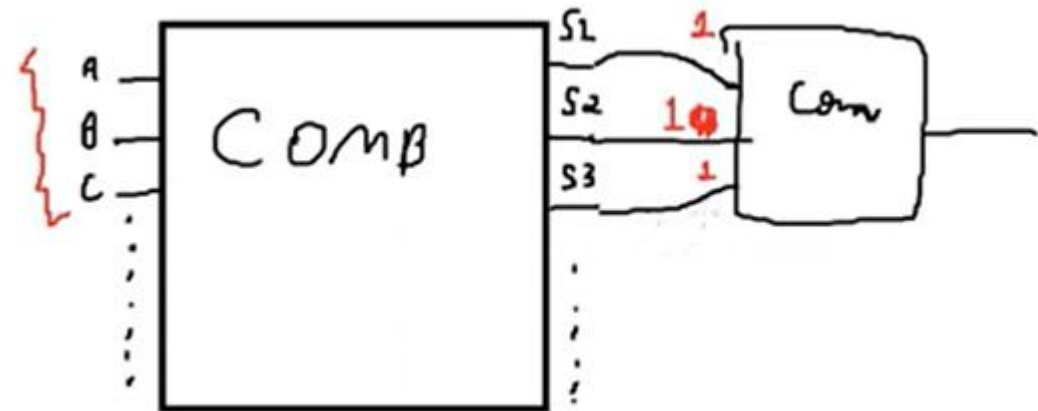
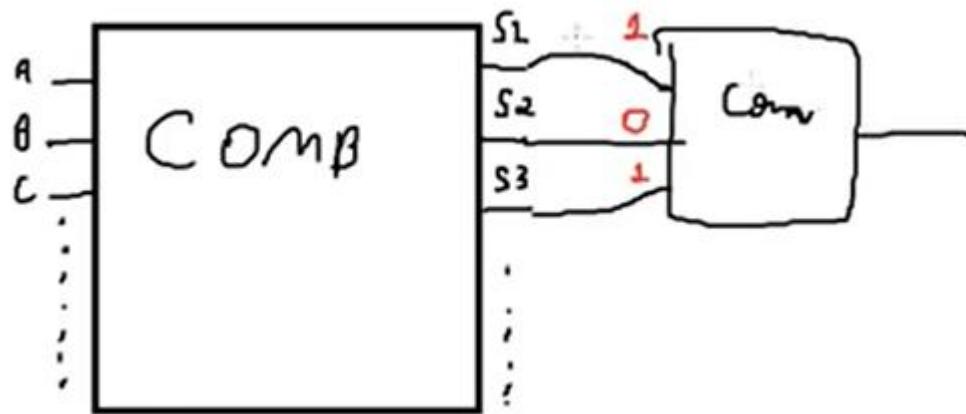


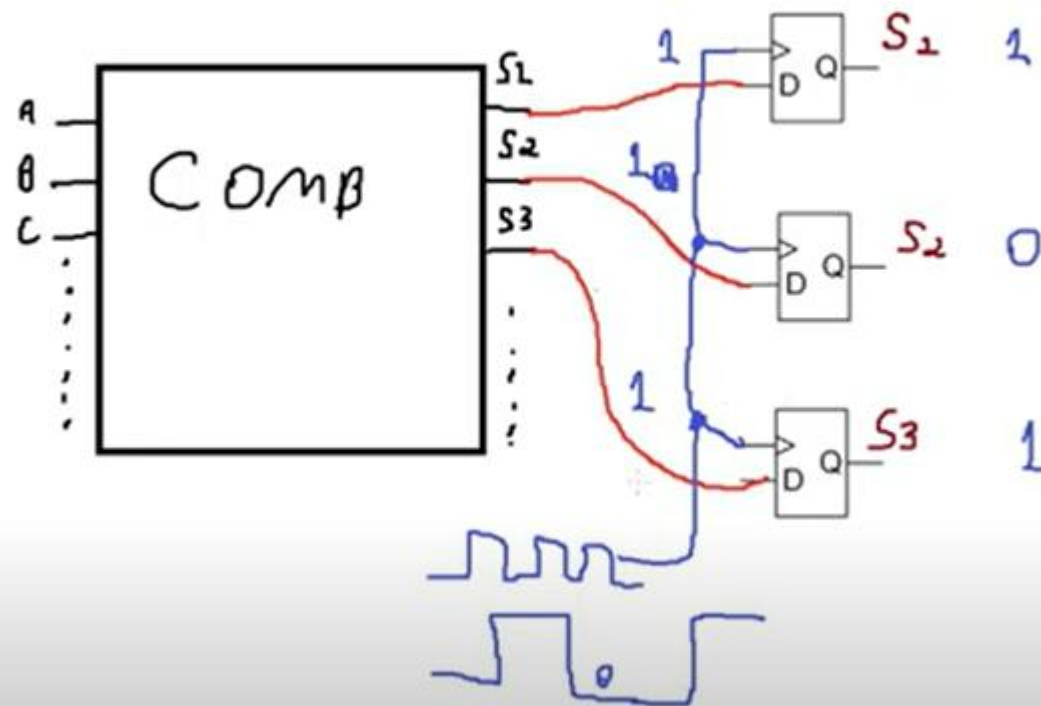
$J = D$	$D = 0$	$D = 1$
$K = \bar{D}$	$J \quad K$	$J \quad K$
	0 1	1 0



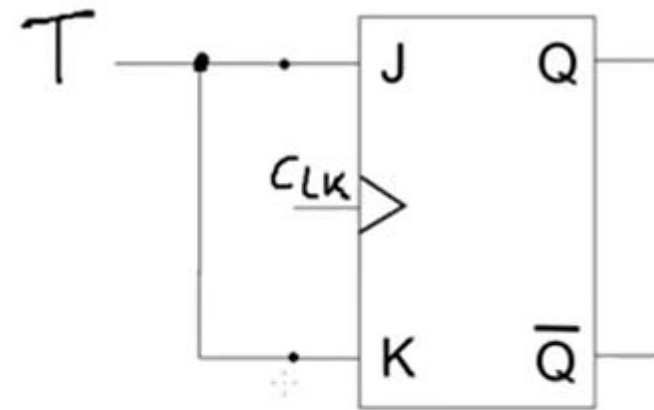
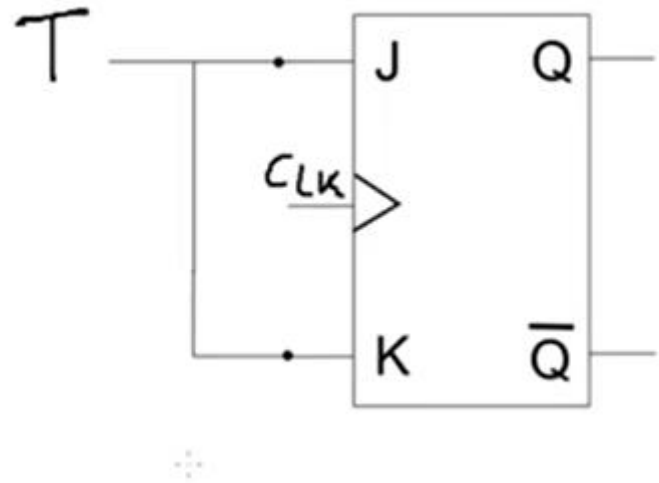
$D = 0$	$D = 1$	D	Q
$Q = 0$	$Q = 1$	0	0
		1	1

Aplicação

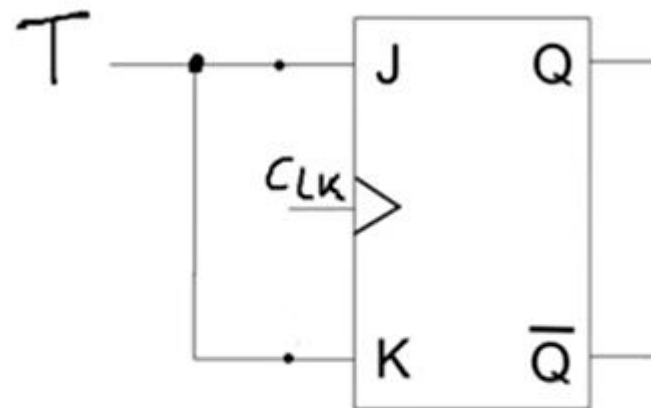




Flip Flop tipo T



$T=0$	$\left. \begin{array}{l} Q \\ \overline{Q} \end{array} \right\}$	$T=1$	$\left. \begin{array}{l} \overline{Q} \\ Q \end{array} \right\}$
$J=0$		$J=1$	
$K=0$		$K=1$	

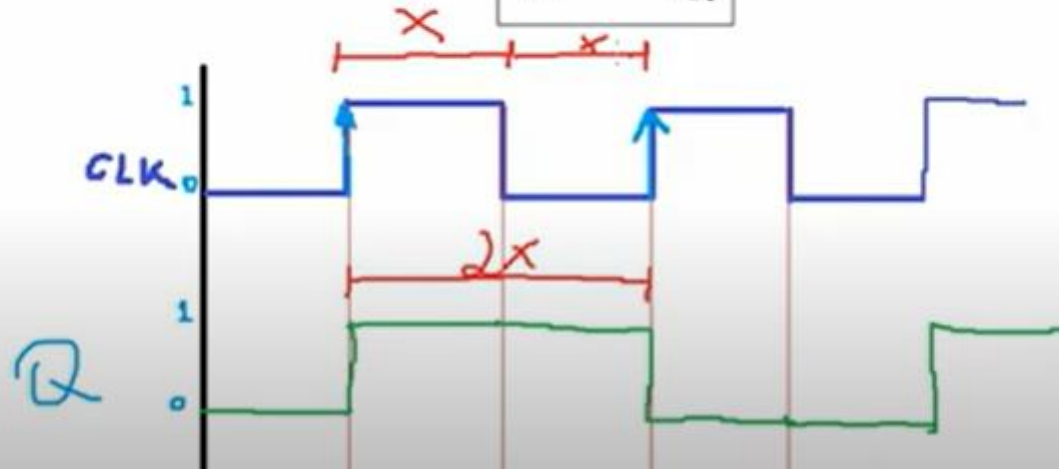
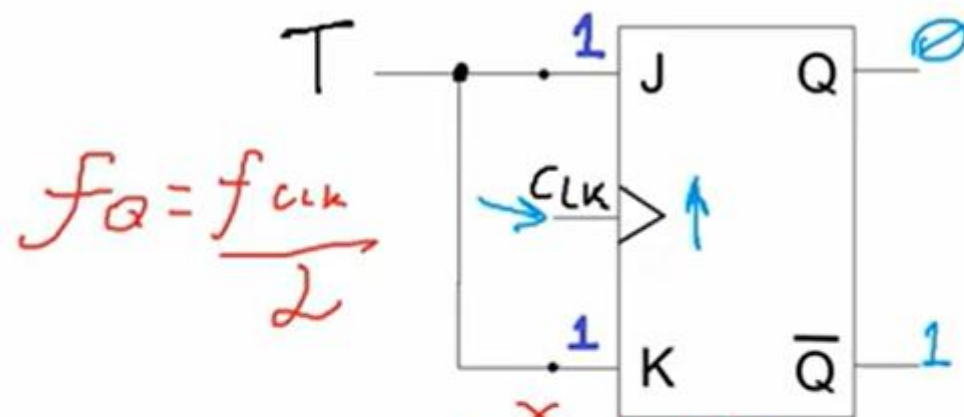


$T=0 \rightarrow Q$ mantém

$T=1 \rightarrow Q$ alternar

Comutador

Alterna na metade frequencia



Registrador de Deslocamento (Conversor)

- Classificação

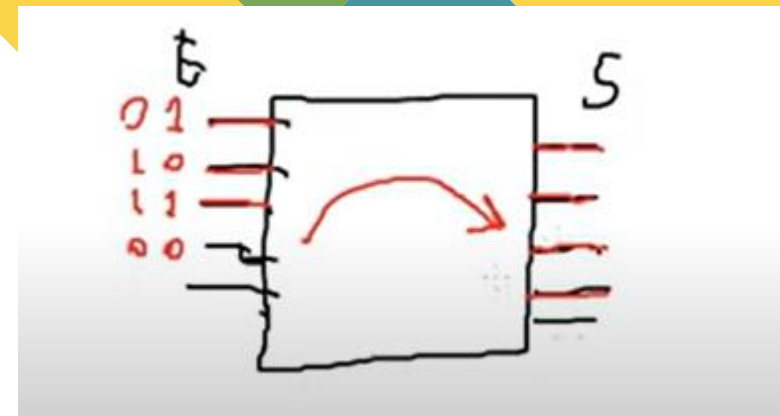
REGISTRADOR DE DESLOCAMENTO

PIPO - PARALLEL INPUT / PARALLEL OUTPUT

SISO - SERIAL INPUT / SERIAL OUTPUT

SIPO - SERIAL INPUT / PARALLEL OUTPUT

PISO - PARALLEL INPUT / SERIAL OUTPUT

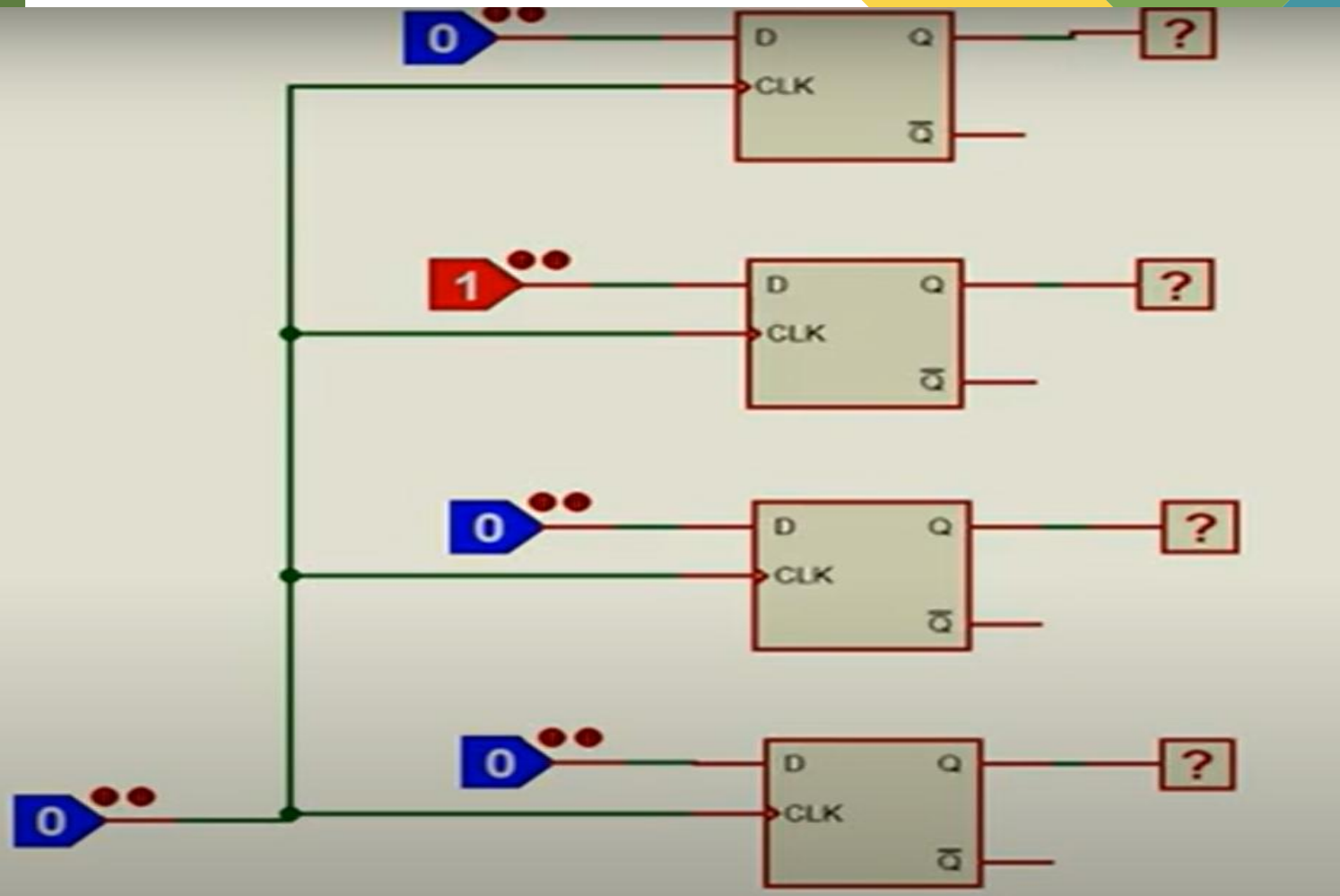


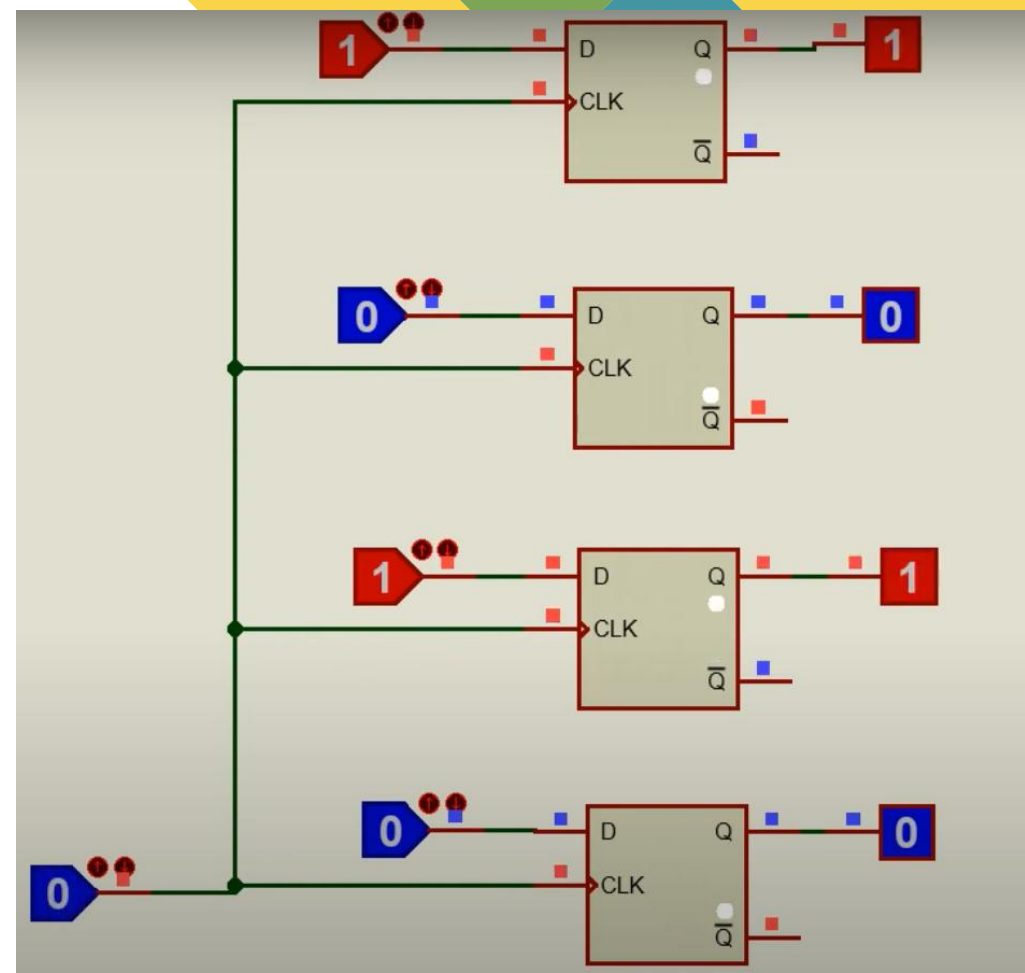
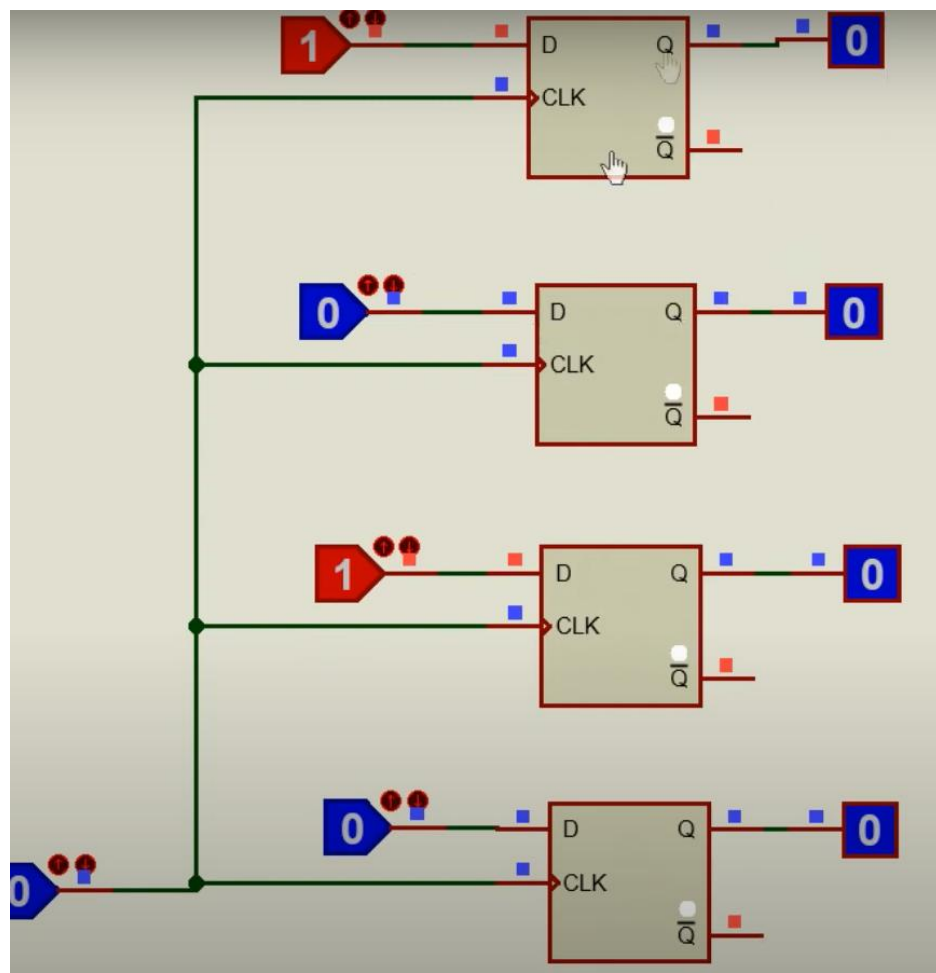
CLK

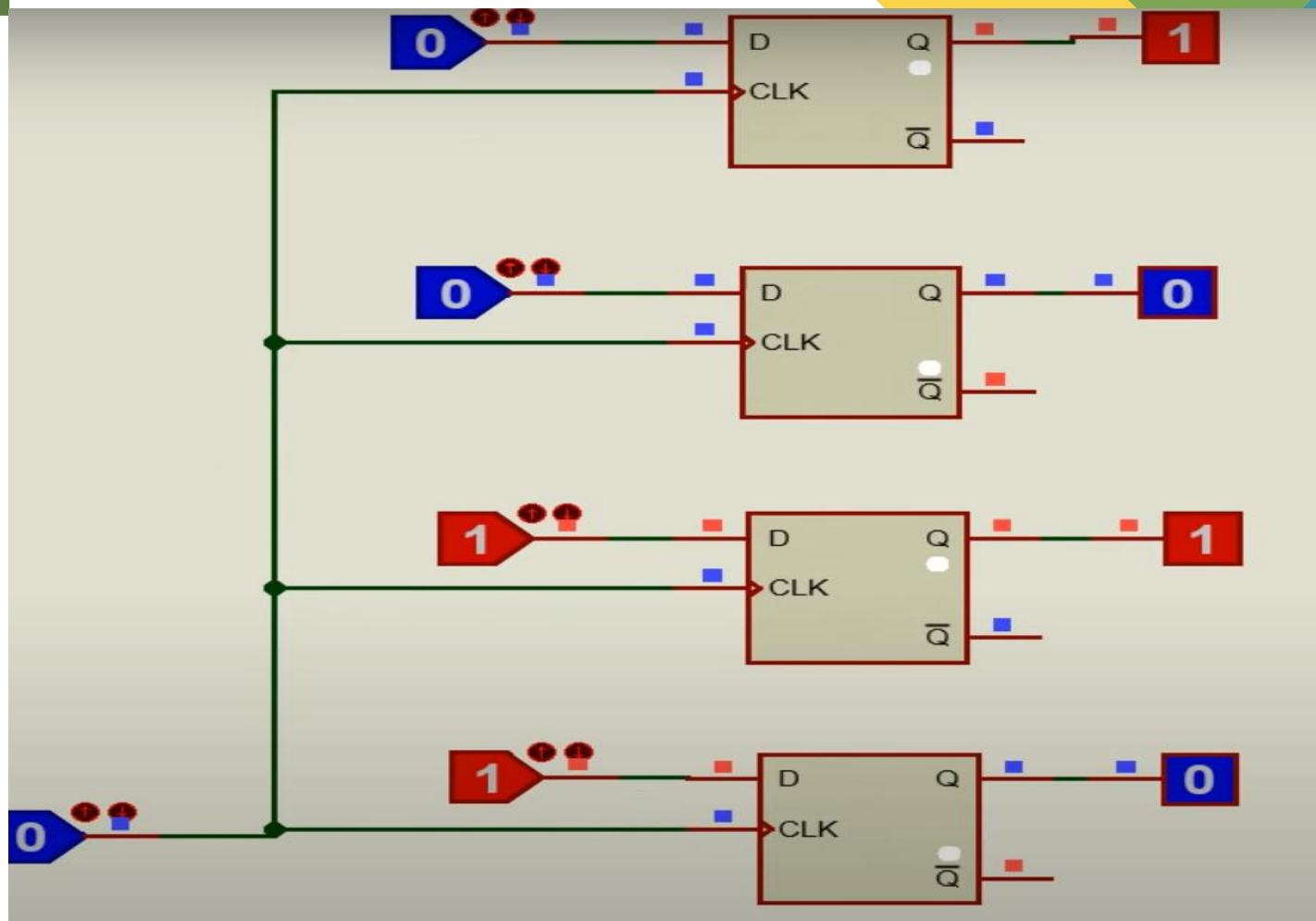
- **Serial**

- Um único meio
- Clock
 - Sinal bem comportado

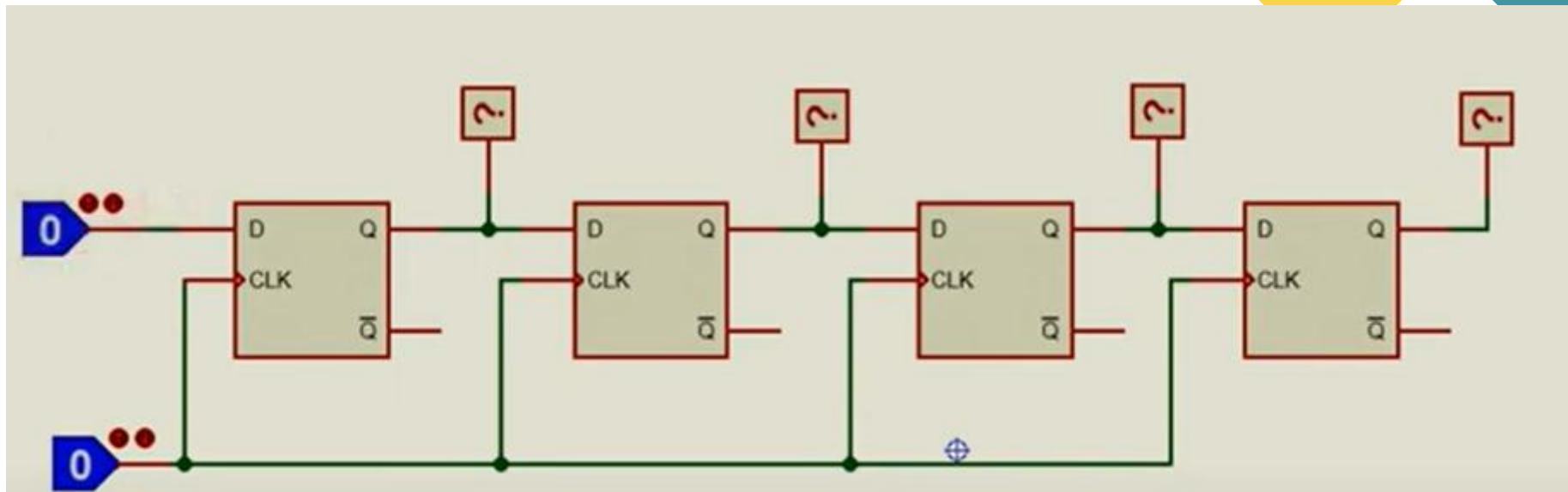
- **Paralelo**







Registrador de deslocamento - SIS0



Saida inicializada

